

C Source Code

```
uint64_t rotate_left(uint64_t x,  
                    uint32_t n) {  
    n &= 63;  
    return (x << n) | (x >> (64 - n));  
}
```

x86-64 (Optimal)

```
movl %esi, %ecx  
movq %rdi, %rax  
rolq %cl, %rax
```

BPF Bytecode

```
r0 = r1  
w2 &= 0x3f  
w1 = 0x40  
w1 -= w2  
r3 = r0  
r3 <=<= r2  
r0 >>= r1  
r0 |= r3
```

x86-64 (Under Optimized)

```
movq %rsi, %rcx  
andq $63, %rcx  
movq %rdx, %r8  
shlq %cl, %r8  
negl %esi  
andq $63, %rsi  
shrxq %rsi, %rdx, %rdx  
orq %rdx, %r8
```

JIT